STRUCTURAL CALCULATION REPORT

Project: Chequered Plate Support Design

Plate Size: 2540 x 375 x 8 mm
Load: 1 Ton (1000 kg) Uniformly Distributed Load
Material: Mild Steel
Yield Strength (Fy): 250 MPa
Modulus of Elasticity (E): 2 x 10^5 MPa
Factor of Safety: 1.5

LOAD CALCULATION

Total Load = 1000 x 9.81 = 9.81 kN
UDL (w) = 9.81 / 2.54 = 3.86 kN/m

PLATE SECTION PROPERTIES

Section Modulus (Z) = bt^2/6
Z = (375 x 8^2) / 6 = 4000 mm^3

Maximum Bending Moment (Full Span)
M = wL^2 / 8
M = 3.86 x 2.54^2 / 8 = 3.11 kNm
M = 3.11 x 10^6 Nmm

Bending Stress = M / Z
= 3.11 x 10^6 / 4000
= 777 MPa (Not Safe for full span)

SAFE SPAN CALCULATION

Allowable Stress = 250 / 1.5 = 150 MPa
Allowable Moment = 150 x 4000 = 0.6 kNm

Safe Span (L)
L = sqrt(8M / w)
L = sqrt(8 x 0.6 / 3.86)
L = 1.12 m

Adopt Support Spacing = 1000 mm
Number of Supports Required = 3 Nos.

C CHANNEL CHECK (100 x 50 x 5 mm)

Assumed Section Modulus Z ≈ 20,000 mm^3
Beam Span = 375 mm

Maximum Moment:
M = wL^2 / 8

= 3.86 x 0.375^2 / 8
= 0.068 kNm
= 68000 Nmm

Bending Stress:
= 68000 / 20000
= 3.4 MPa (Safe)

WELD DESIGN

Reaction per support = 9.81 / 3 = 3.27 kN
Adopt 4 kN including impact

Required weld area = 4000 / 100 = 40 mm^2

4 mm fillet weld capacity:
Effective throat = 0.7 x 4 = 2.8 mm
Provided area (50 mm length) = 2.8 x 50 = 140 mm^2 (Safe)

FINAL DESIGN SUMMARY

Plate: Safe at 1000 mm support spacing
C Channel 100x50x5 mm: Safe
4 mm Fillet Weld: Safe
Deflection: Within permissible limit

End of Report